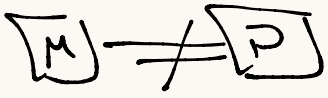
system bus



system bust

↳ it's wai/ to Communicates systemParts tosether

↳ type → wire → white board
↳ track → Pcb
↳ tunnal → chip

↳ system bus Content →
↳ ① data bus
↳ ② Address bus
↳ ③ Control bus

↳ system bus → ≠ ⇒

if You have system bus = 32 bit

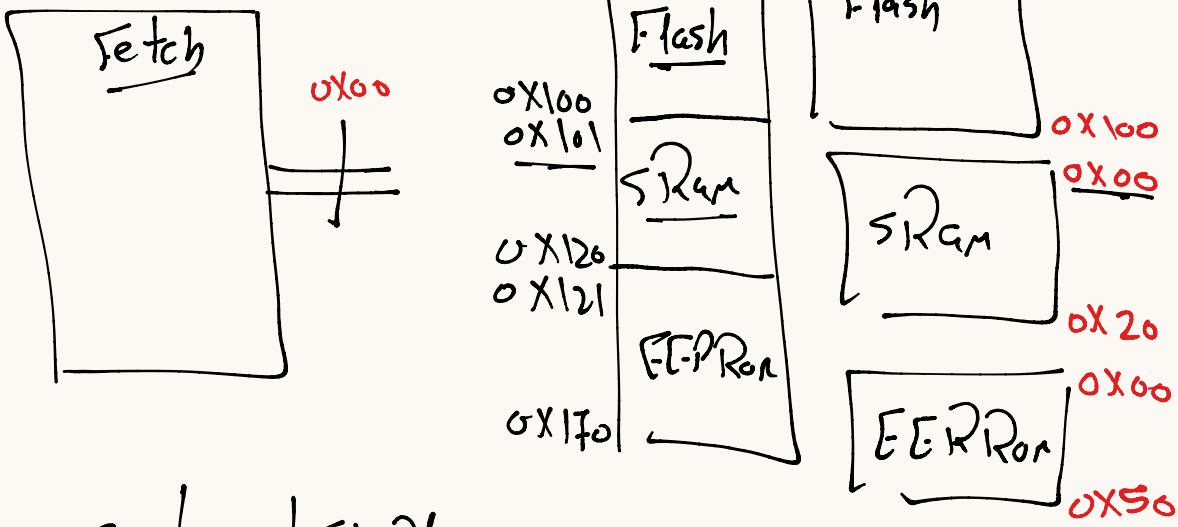
↳ data bus = 32 bit = 32 wire
↳ var ⇒ 4byte

↳ Address bus = 32 bit ⇒ 32 wire
↳ Address → 32 bit

NOTE IN Created by Notein

System Type Arch

① Von neumann

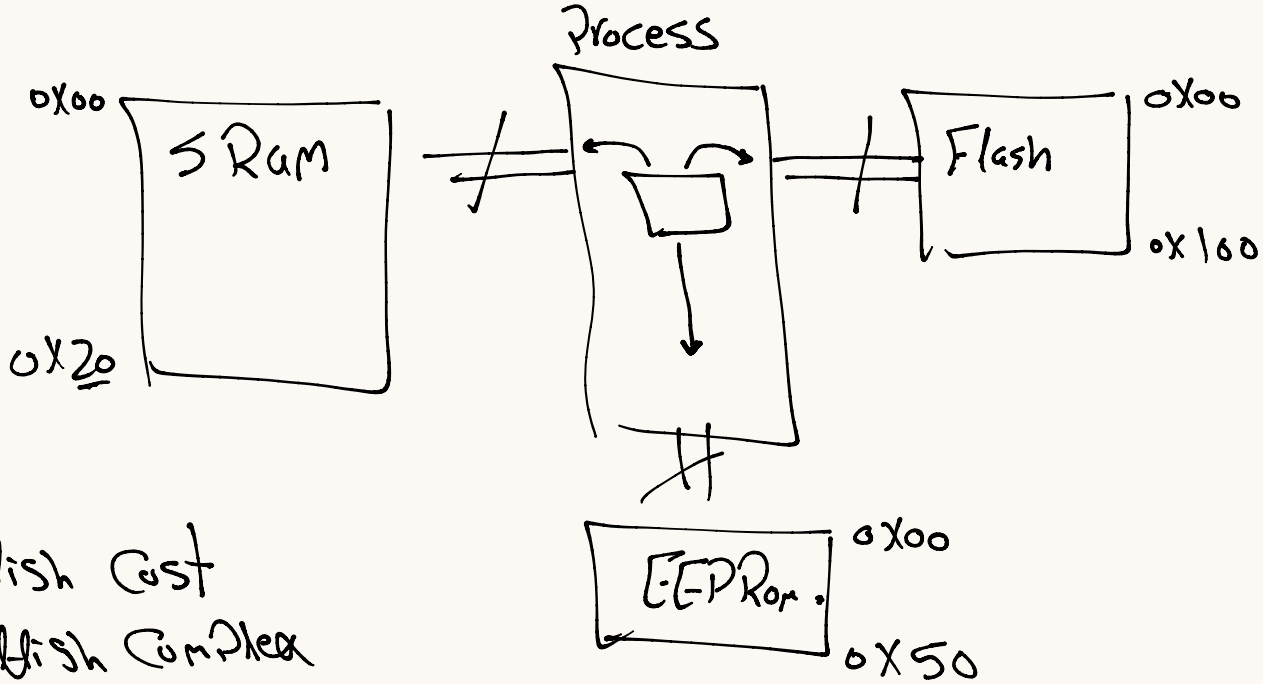


↳ low cost | simple

↳ take long time

---

② Harvard Arch :. For each memory → system Bus



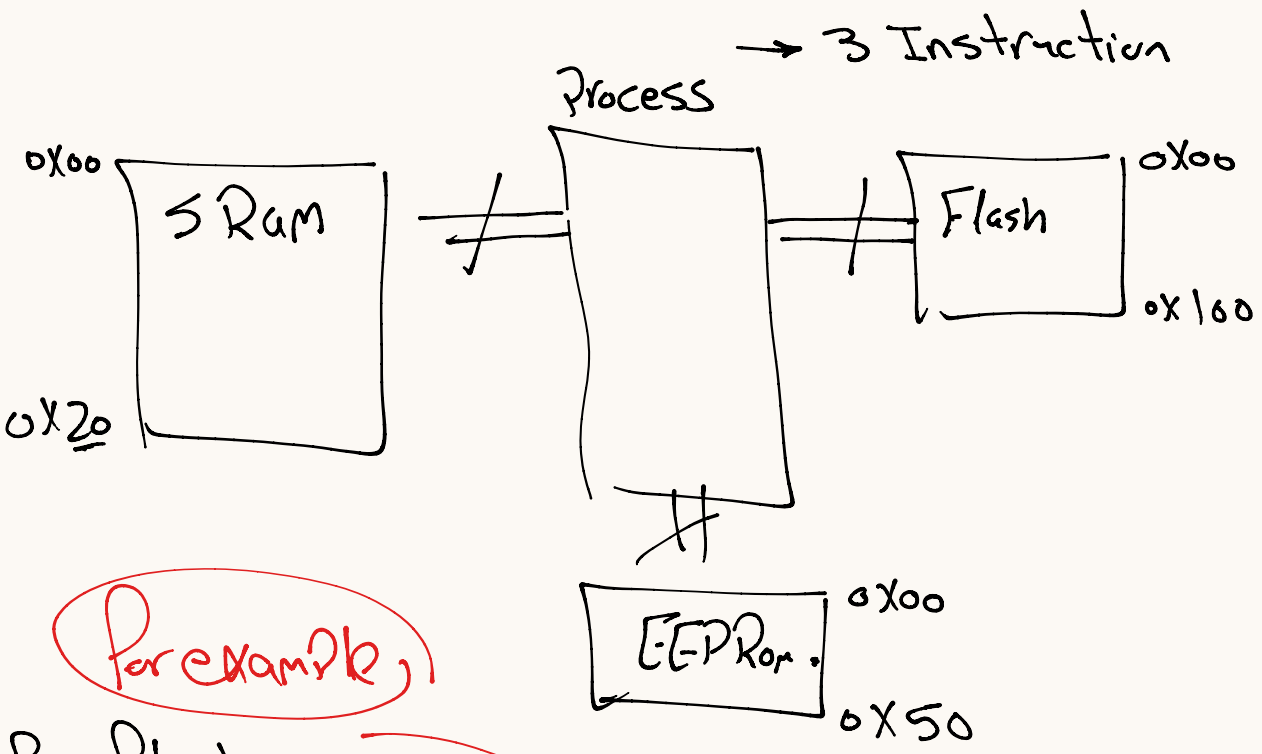
↳① High Cost

↳② High Complex

---

ex:- `*((char *)0x20) = 20;`

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for example,

→ For Flash

↳ Assembly Instruction ⇒ store / load

→ For SRam

↳ Assembly Instruction ⇒ Read / write

→ for EEPRom

↳ Assembly Instruction ⇒ In / out

---

if we write Code C → Compiler Convert for which Instruction

Compiler by default will → Convert to SRam Instruction

if you need to write EEPROM → Must be write Assembly

extern .asm ↳

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Von numman

→ Code_C
→ take long time

Harvard

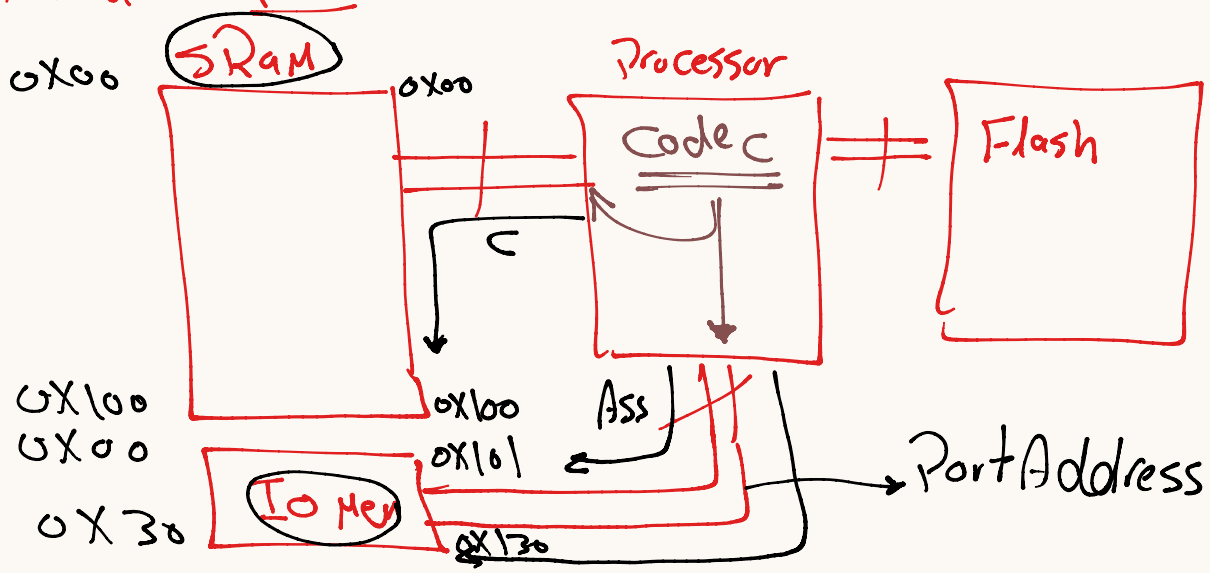
→ Code C for SRAM
otherwise → write Assembly
→ Fast time

Memory map

Port map

↳ Foreach system bus have Instruction.

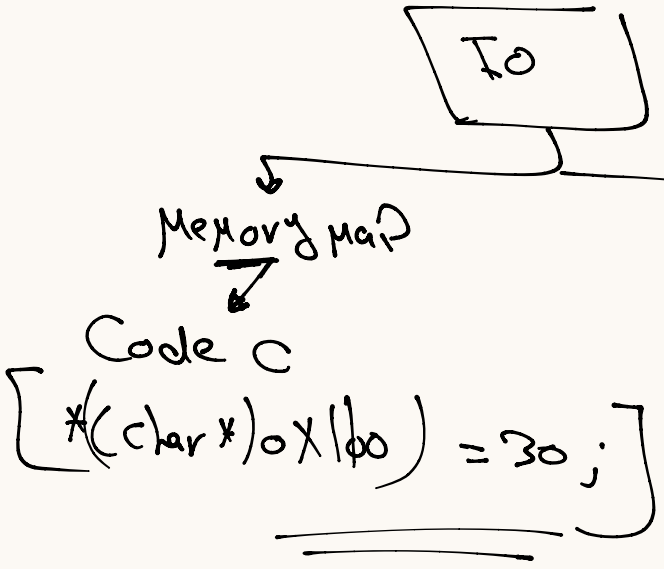
* Hybrid → AVR



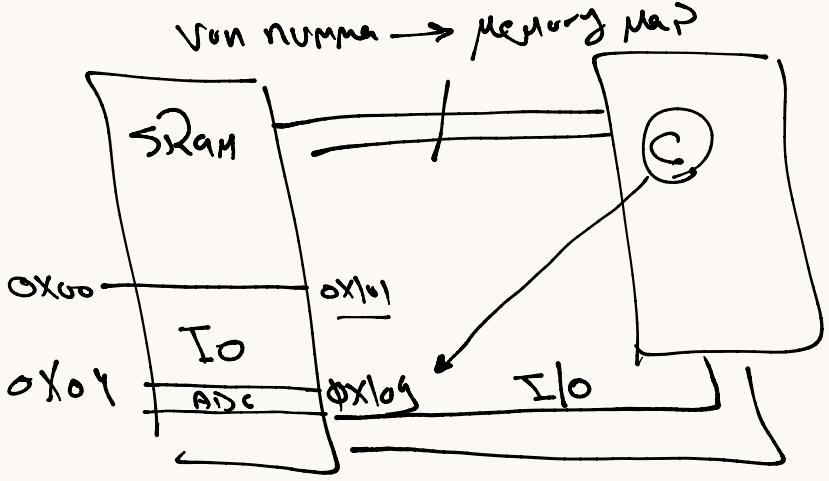
write C_Code to Io_Memory

Io Memory

Port map (
Assembly



NOTE IN Created by Notein



Von numman

↙

* Memory Map

* C_ Code

Harvard

↙

* Port Map → 0x04

* Assembly

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